

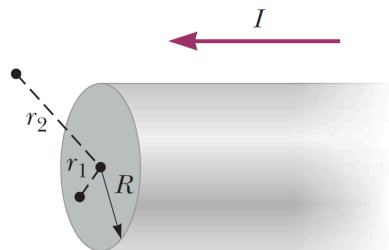
7. class, Electrodynamics and Optics

Page 792, Task 22.

A long, cylindrical conductor of radius R carries a current I as shown in the figure. The current density J , however, is not uniform over the cross section of the conductor but rather is a function of the radius according to $J = br$, where b is a constant. Find an expression for the magnetic field magnitude B

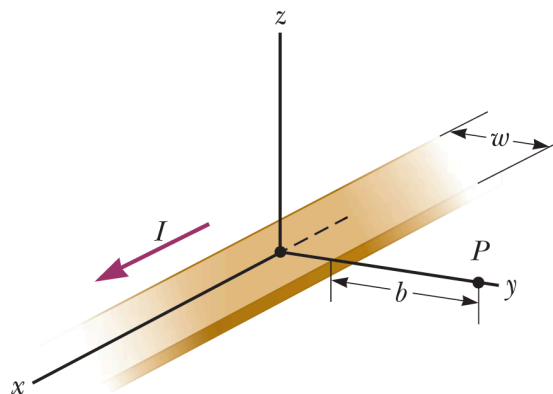
(a) at a distance $r_1 < R$

(b) at a distance $r_2 > R$, measured from the center of the conductor.



Page 794, Task 34.

A very long, thin strip of metal of width w carries a current I along its length as shown in the figure. The current is distributed uniformly across the width of the strip. Find the magnetic field at point P in the diagram. Point P is in the plane of the strip at distance b away from its edge.



Page 794, Task 36.

A nonconducting ring of radius R is uniformly charged with a total positive charge q . The ring rotates at a constant angular speed ω about an axis through its center, perpendicular to the plane of the ring. What is the magnitude of the magnetic field on the axis of the ring a distance $\frac{1}{2}R$ from its center?

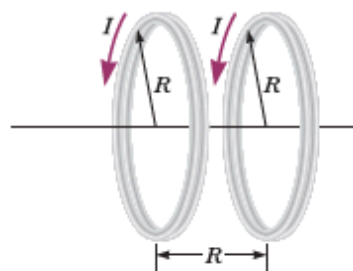
Page 794, Task 38.

Two circular coils of radius R , each with N turns, are perpendicular to a common axis. The coil centers are a distance R apart. Each coil carries a steady current I in the same direction as shown in the figure.

(a) Show that the magnetic field on the axis at a distance x from the center of one coil is

$$B = \frac{N\mu_0 IR^2}{2} \left[\frac{1}{(R^2 + x^2)^{3/2}} + \frac{1}{(2R^2 + x^2 - 2Rx)^{3/2}} \right]$$

(b) Show that dB/dx and d^2B/dx^2 are both zero at the point midway between the coils. We may then conclude that the magnetic field in the region midway between the coils is uniform. Coils in this configuration are called *Helmholtz coils*.



Page 794, Task 37.

A very large parallel-plate capacitor has uniform charge per unit area $+\sigma$ on the upper plate and $-\sigma$ on the lower plate. The plates are horizontal, and both move horizontally with speed v to the right.

- What is the magnetic field between the plates?
- What is the magnetic field just above or just below the plates?
- What are the magnitude and direction of the magnetic force per unit area on the upper plate?
- At what extrapolated speed v will the magnetic force on a plate balance the electric force on the plate?

Suggestion: Use Ampere's law and choose a path that closes between the plates of the capacitor.

Page 796, Task 49.

A long, cylindrical conductor of radius a has two cylindrical cavities each of diameter a through its entire length as shown in the end view of the figure. A current I is directed out of the page and is uniform through a cross section of the conducting material.

Find the magnitude and direction of the magnetic field in terms of μ_0 , I , r and a at (a) point P_1 and (b) point P_2 .

